

5.1.4 Components of Neural Networks

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوشیدہ تہ، وزن (Weights)، بائیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 05 of 15

★ Golden

5.1.4 Components of Neural Networks ★ Golden

LEARN IT

The components of a neural network can be understood through its **layered architecture**. Data **enters**, gets **processed**, and **exits**, with mathematical adjustments happening inside.

#	Component	Description
1	Input layer	Receives the initial raw data (for example image pixels, sound waves, text).
2	Neurons / nodes	Small processing units that receive, calculate, and pass information forward.
3	Hidden layer(s)	The layers between input and output. They process the data and learn patterns from it.
4	Weights	Numbers that show the importance or strength of connections between neurons.
5	Bias	An extra constant value added to help the model adjust its result and improve accuracy.
6	Activation function	A mathematical gate that decides whether the information is important enough to move to the next layer.

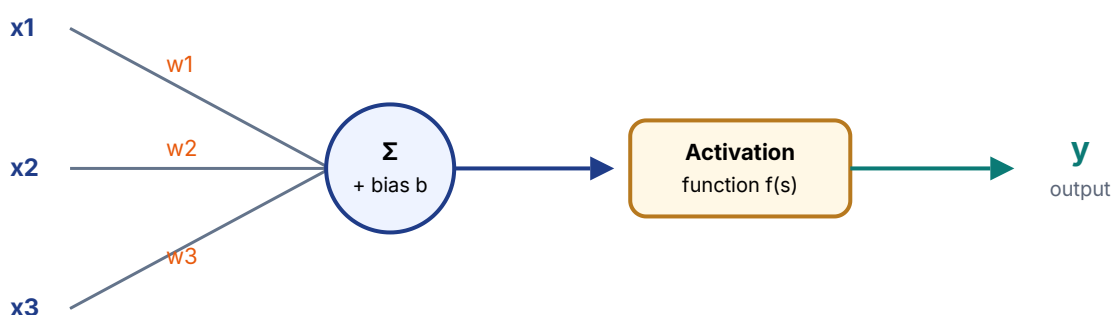
7	Output layer	Gives the final computed result, decision or prediction (for example "This is a cat").
8	Loss function	Checks the error (how much the model's answer is wrong compared with reality) so the network can correct itself.

What happens inside one neuron

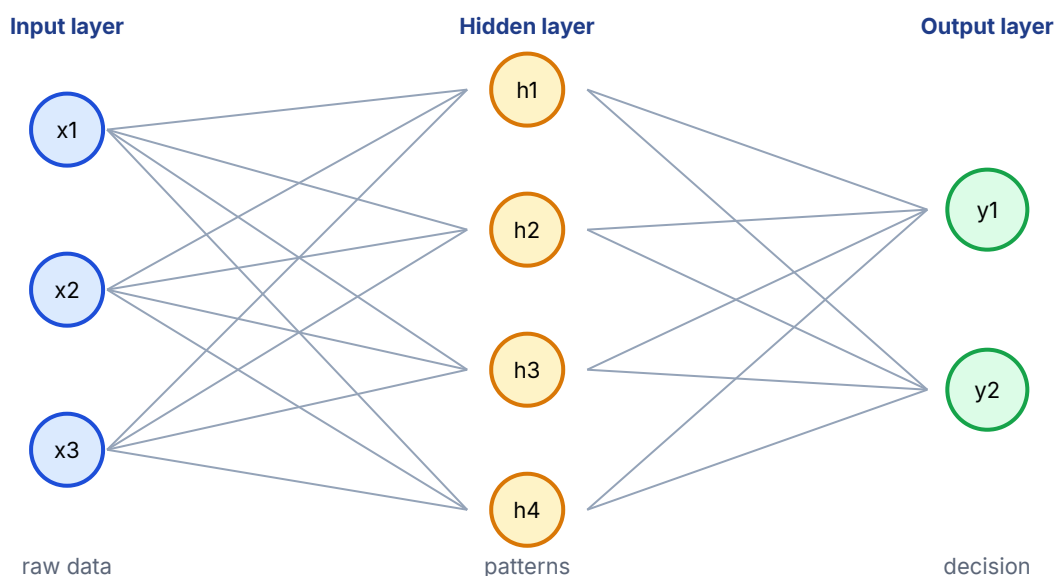
Each neuron does a short calculation:

1. Multiply each input x by its weight w .
2. Add those products together, then add the **bias** b . This total is the **weighted sum** s .
3. Pass s through an **activation function**. A simple exam-friendly gate is a **threshold (step) function**: if s is at least the threshold, the neuron **fires** (output = 1); otherwise output = 0.

In symbols: $s = (x_1 \times w_1) + (x_2 \times w_2) + \dots + b$, then $y = 1$ if $s \geq \text{threshold}$, else $y = 0$.



One neuron: inputs $\times$ weights are summed, bias is added, then an activation function decides the output y .



A small neural network: 3 input nodes, 4 hidden nodes, 2 output nodes. Each line is a weighted connection.

WORKED EXAMPLE

Worked example: neuron calculation with numbers. A college in Larkana uses a tiny network to guess whether a student is ready for a test. One hidden neuron receives two inputs:

Symbol	Meaning	Value
x1	Hours studied yesterday (scaled 0 to 1)	0.8
x2	Attendance this month (scaled 0 to 1)	0.5
w1	Weight of hours studied	0.4
w2	Weight of attendance	0.6
b	Bias	0.1
threshold	Activation gate	0.5

Step 1 — weighted products.

$$x1 \times w1 = 0.8 \times 0.4 = \mathbf{0.32}$$

$$x2 \times w2 = 0.5 \times 0.6 = \mathbf{0.30}$$

Step 2 — add bias to get the weighted sum s.

$$s = 0.32 + 0.30 + 0.1 = \mathbf{0.72}$$

Step 3 — activation (threshold 0.5). Is $0.72 \geq 0.5$? **Yes**, so the neuron fires: **y = 1** (ready for the test).

Step 4 — a second student with $x1 = 0.2$ (almost no study) and $x2 = 0.3$:

$$s = (0.2 \times 0.4) + (0.3 \times 0.6) + 0.1 = 0.08 + 0.18 + 0.1 = \mathbf{0.36}$$

0.36 is **less than** 0.5, so **y = 0** (not ready). The same neuron, different inputs, different decision.

You can check the first student in Python 3:

```
x1, x2 = 0.8, 0.5
w1, w2 = 0.4, 0.6
b = 0.1
s = x1 * w1 + x2 * w2 + b
y = 1 if s >= 0.5 else 0
print(round(s, 2))
print(y)
```

OUTPUT

0.72

1

Loss function, briefly. If the true answer for the first student was actually "not ready" (true $y = 0$) but the model said 1, the loss is high. Training then slightly changes weights and bias so the next guess is closer to reality.

REMEMBER IT

Weights = importance. Bias = extra nudge. Activation = yes/no gate. Loss = how wrong. A large weight means that input matters more. Bias shifts the sum up or down so the gate can open at the right place.

COMMON MISTAKE

Mistake: Forgetting to add the bias, or multiplying instead of adding after the products.

Fix: Always **multiply first** (each x by its w), **then add** the products, **then add b** , **then** apply the activation. Write the four lines in the exam as we did above.

اردو میں سمجھیں

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوشیدہ تہ، وزن (Weights)، بائیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔ ہر نیورون ان پٹ کو وزن سے ضرب دے کر جمع کرتا ہے، بائیس جوڑتا ہے، پھر ایک گیٹ فیصلہ کرتا ہے کہ سگنل آگے جائے یا نہیں۔ وزن تعلق کی طاقت بتاتا ہے؛ لاس فنکشن غلطی ناپتا ہے تاکہ ماڈل خود درست ہو سکے۔

EXAM TIP

Golden questions: "List the components of a neural network" (often 8 names + one line each), "What are weights and bias?", and a **numerical neuron**: given x , w , b and a threshold, find s and y .

Show every arithmetic step. Underline the final y . If asked for a diagram, draw Input → Hidden → Output with circles and connecting lines, and label weights on the lines.

Mention the loss function as the component that measures error so the network can correct itself.

CHECK YOURSELF

1. What do weights show?
2. Using $x_1 = 1$, $x_2 = 0$, $w_1 = 0.5$, $w_2 = 0.5$, $b = 0$, threshold = 0.4, what is y ?
3. What does the loss function check?

Answers: 1. The importance or strength of connections between neurons. 2. $s = 1 \times 0.5 + 0 \times 0.5 + 0 = 0.5$; $0.5 \geq 0.4$ so $y = 1$. 3. The error — how much the model's answer is wrong compared with reality.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Input layer:

- a. Receives the features
- b. Always outputs the class
- c. Stores SQL
- d. Draws Figma

MCQ 2. Hidden layers:

- a. Transform features (learned)
- b. Are optional pie charts
- c. Are OSI hubs
- d. Are primary keys

MCQ 3. Output layer:

- a. Produces the prediction
- b. Reads the keyboard
- c. Commits sqlite
- d. Opens VS Code

MCQ 4. Weights are:

- a. Strengths of connections, learned from data
- b. Font weights only
- c. TCP weights
- d. Agile story points

MCQ 5. Activation functions:

- a. Add non-linearity so layers can do more than a straight line
- b. Sort the array
- c. Encrypt the disk
- d. Draw ER

MCQ 6. This is ★: a 5-mark wants:

- a. Named parts + one sentence each + tiny diagram
- b. Only the word "brain"
- c. Only careers
- d. Only MVPs

Extra short questions

1. List components of a NN from the notes.
2. What does a neuron do conceptually?
3. Why more than one hidden layer can help.
4. What is not a component?

Extra long question

1. Draw and label input, 2 hidden, output. Annotate weights on two edges. Explain in 4 sentences how a pass/fail signal moves left to right. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Layers (in/hidden/out), neurons/nodes, weights, activations, (bias).
2. Weighted sum of inputs, then activation.
3. Features of features (deep).
4. A pie chart; a primary key; a sprint.

Long

1. Diagram. Forward pass description. Training mentioned as changing weights, not needed in full backprop maths. One misuse: calling it an exact brain.

Next lecture: 5.1.5 Components of Deep Learning